

DSBDAL Assignment 12

Write a simple program in SCALA using Apache Spark framework

Steps for installation of Scala and Apache Spark-

1. Download latest version of JDK from <https://www.oracle.com/java/technologies/downloads/#java8-windows>
2. Download and execute scala installer from <https://www.scala-lang.org/download/>
3. Launch scala-ammonite using
cs launch ammonite --scala 2.13
4. Download Apache Spark from <https://spark.apache.org/downloads.html> and extract the folder in C drive.
5. Download hadoop.dll and winutils.exe from <https://github.com/kontext-tech/winutils/blob/master/hadoop-3.3.0/bin/hadoop.dll>
6. Create a folder on C drive named as 'Winutils' and then create a folder for 'bin' and place hadoop.dll and winutils.exe there.
7. Set up environmental variables name=SPARK_HOME and value=path of spark folder
8. Set up environmental variables name=Winutils and value=path of winutils folder
9. Set up the path->new-
%WINUTILS%\bin
%SPARK_HOME%\bin
10. Open cmd and redirect to bin folder cd spark
11. Type spark-shell or pyspark to start spark framework

WordCount.scala-Save the file C->spark-proj->WordCount.scala

```
import org.apache.spark.sql.SparkSession
```

```
object WordCount {  
  def main(args: Array[String]): Unit = {  
  
    val spark = SparkSession.builder()  
      .appName("Word Count Example")  
      .master("local[*]")  
      .getOrCreate()  
  
    val sc = spark.sparkContext  
    val data = sc.textFile("C:\\spark-proj\\input.txt")  
  
    val counts = data  
      .flatMap(_.split(" "))  
      .map((_,1))
```

```

        .reduceByKey(_+_ )

counts.collect().foreach(println)

spark.stop()
}
}

```

input.txt-Save the file C->spark-proj->input.txt

AJK PPS

AJK

DP

SD

For execution using Apache Spark-

1. Open cmd in administrator mode
2. Redirect to c->spark-proj
3. Type spark-shell
4. Spark environment will open
5. Type :load WordCount.scala
6. Type WordCount.main(Array())
7. Check the output

Output:

```

C:\Windows\System32>cd \
C:\>cd spark-proj
C:\spark-proj>spark-shell
WARNING: Using incubator modules: jdk.incubator.vector
26/04/20 09:43:14 WARN Shell: Did not find winutils.exe: java.io.FileNotFoundException: java.io.FileNotFoundException: HADOOP
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to

  ____      __
 / ___ |__ /  \
| |  \|_  \_  /
| |___| |_) |
|_____|____|
version 4.1.1

Using Scala version 2.13.17 (OpenJDK 64-Bit Server VM, Java 17.0.18)
Type in expressions to have them evaluated.
Type :help for more information.
Spark context Web UI available at http://DESKTOP-9HRLKUN:4040
Spark context available as 'sc' (master = local[*], app id = local-1776658398196).
Spark session available as 'spark'.

scala> :load WordCount.scala
val args: Array[String] = Array()
Loading WordCount.scala...
import org.apache.spark.sql.SparkSession
object WordCount

scala> WordCount.main(Array())
26/04/20 09:43:46 WARN SparkSession: Using an existing Spark session; only runtime SQL configurations will take effect.
(DP,1)
(AJK,2)
(SD,1)
(PPS,1)

scala>

```

DSBDAL Assignment 13

Locate dataset (e.g., sample_weather.txt) for working on weather data which reads the text input files and finds average for temperature, dew point and wind speed

```
In [1]: import csv

temp_sum = dew_sum = wind_sum = count = 0

with open(r"C:\Users\STUDENT\Desktop\weather_project\sample_weathertxt.txt",
          "r") as file:
    reader = csv.reader(file)
    next(reader)

    for row in reader:
        try:
            temp = float(row[1])
            dew = float(row[2])
            wind = float(row[3])

            temp_sum += temp
            dew_sum += dew
            wind_sum += wind
            count += 1
        except:
            continue

print("Avg Temperature:", temp_sum / count)
print("Avg Dew Point:", dew_sum / count)
print("Avg Wind Speed:", wind_sum / count)
```

Avg Temperature: 25.85

Avg Dew Point: 18.375

Avg Wind Speed: 12.0

Date	Temp(C)	DewPoint(C)	WindSpeed(km/h)
2024-01-01	25	18	12
2024-01-02	27	19	15
2024-01-03	26	20	10
2024-01-04	28	21	14
2024-01-05	29	22	13
2024-01-06	30	23	16
2024-01-07	31	24	18

Save this as:

sample_weather.txt